

# Enhancing Visuo-Haptic Coherency by Manipulating Fingertip Contact Tilt

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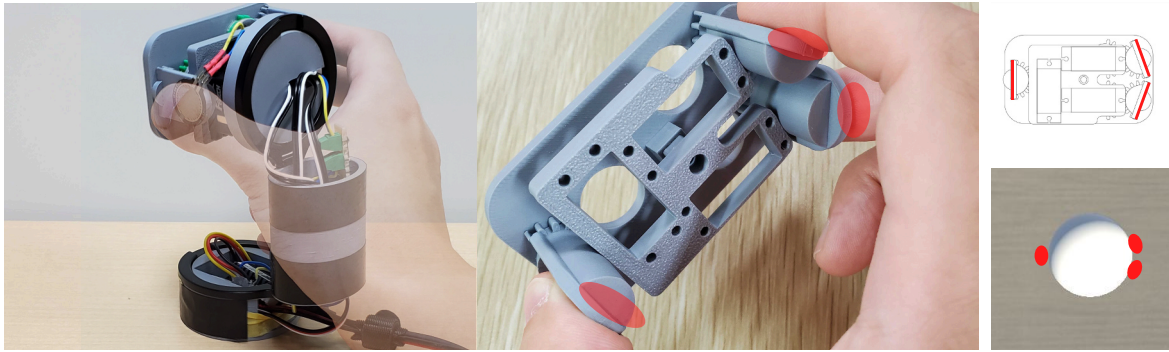


Figure 1: A shape presentation device that manipulates the inclination of fingertip contact planes.

## ABSTRACT

This study investigates the manipulation of fingertip contact plane manipulation to enhance visuo-haptic coherency of virtual object. We designed a compact device that adjusts fingertip contact plane tilt to simulate diverse 3D shapes. Through user studies, the device demonstrated significant improvements in haptic realism, enjoyment, and operability compared to conventional methods. These findings highlight the potential of fingertip contact plane manipulation as a novel approach for advancing virtual reality (VR) haptic interfaces, enabling more immersive and realistic experiences.

**Index Terms:** Shape, Grasping interaction, Handheld device

## 1 INTRODUCTION

Immersive VR experiences with Head-Mounted Displays (HMD) allow users to effortlessly immerse themselves in diverse environments for gaming or work by displaying objects with varied appearances. However, while visual experiences can be freely edited, haptic experiences lack such flexibility. Regardless of the objects users grasp within the VR environment, the tactile sensations are limited to the cylindrical controllers or the softness of their fingertips. A device capable of dynamically presenting object shapes is necessary to achieve grasping and manipulation experiences in VR that are equivalent to those in the real world.

To make VR experiences closer to real-world interactions, gloves that restrict hand movements [3] and VR controllers with variable shapes [5, 2] have been researched and developed. These devices primarily focus on the positions of users' fingertips, fixing them to match the contours of virtual objects to present diverse shapes.

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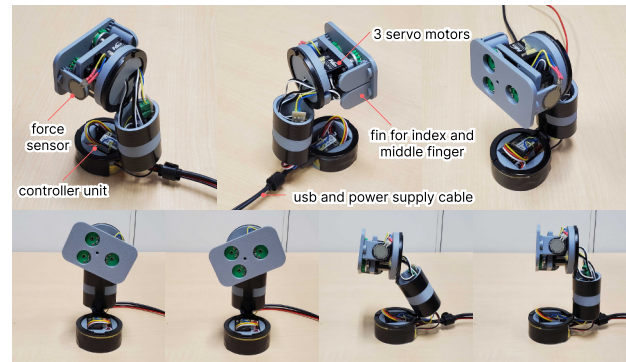


Figure 2: Overview of device and its components.

However, most of these methods concentrate on fingertip positions and forces, neglecting the inclination of the surfaces that the fingertips contact. Several studies have shown that the proprioceptive sense of fingertip position can be manipulated through visual information [1]. Furthermore, in the perception of curvature, the normal direction of the surface contacted by each fingertip plays a more crucial role than the fingertip positions themselves [4]. Therefore, to enhance VR experiences to be more realistic, a shape presentation device that focuses on the directions of the planes contacted by each fingertip could be beneficial.

## 2 DEVICE DESCRIPTION

The device used in this study is a handheld haptic interface designed to manipulate the inclination of fingertip contact planes. The device consists of three servo motors, each connected to fins that make contact with the thumb, index finger, and middle finger. By rotating the fins within a range of  $\pm 45^\circ$ , the device simulates the tilt of contact planes, enabling the perception of various 3D shapes.

Servo motors operate under a Proportional-Integral-Derivative (PID) control loop, allowing precise angle adjustments. Communication between the device and the PC is facilitated via USB and wireless protocols, ensuring real-time control and data transmis-

sion. The overall dimensions of the device are compact ( $140 \times 90 \times 70$  mm) and portable, making it suitable for diverse VR scenarios.

### 3 USER STUDY: VISUO-HAPTIC COHERENCY

#### 3.1 Experiment Description

We conducted a study to evaluate whether fingertip contact plane manipulation enhances the congruence between visual and haptic perceptions. The hypotheses were:

- H1: Contact plane tilt affects visuo-haptic coherency.
- H2: Manipulating fingertip contact planes improves the quality of VR grasping experiences.

#### 3.2 Participants

Eighteen participants (9 male, 9 female, mean age 23.8 years, SD = 3.68) with an average hand length of 18.06 cm (SD = 1.16 cm) participated. Ethical approval was obtained prior to the study.

#### 3.3 Experimental Setup and Procedure

Participants used three methods to manipulate virtual objects: a standard VR controller, hand tracking, and a device enabling fingertip contact plane manipulation. Each condition involved grasping eight predefined shapes, followed by subjective evaluations on visual-haptic coherency, fatigue, enjoyment, and operability. Data were collected using a 7-point Likert scale and a visual analog scale (VAS).

#### 3.4 Results

##### 3.4.1 Visual and Haptic Coherency by Shape

A Shapiro-Wilk test was conducted for the shape factor, and normality was not confirmed for the `cylinder_SIDE` shape and the cube shape (`cylinder_SIDE`:  $p = 0.027$ , cube:  $p = 0.03$ ). Therefore, a non-parametric Friedman test was performed, which did not detect significant differences ( $p = 0.66$ , statistic = 2.43,  $W = 0.01$ ).

##### 3.4.2 Visual and Haptic Coherency by Operation Method

A Shapiro-Wilk test was conducted on the operation method factor, and normality was not confirmed for any methods (con:  $p < 0.001$ , han:  $p < 0.001$ , fre:  $p < 0.001$ ). A Friedman test detected significant differences ( $p < 0.001$ , statistic = 16.8,  $W = 0.47$ ). Post-hoc Wilcoxon signed-rank tests revealed significant differences between con and fre conditions ( $p < 0.001$ , statistic = 6.0,  $r = 0.92$ ) and between han and fre conditions ( $p < 0.001$ , statistic = 5.0,  $r = 0.62$ ).

##### 3.4.3 Subjective Evaluations

A Shapiro-Wilk test was conducted for each subjective metric, and normality was not confirmed for Fatigue (con:  $p = 0.003$ , han:  $p < 0.001$ , fre:  $p = 0.0016$ ), Enjoyment (fre:  $p < 0.001$ ), and Operability (con:  $p = 0.044$ , han:  $p = 0.018$ , fre:  $p = 0.003$ ). A Friedman test detected significant differences in Enjoyment ( $p = 8.03 \times 10^{-6}$ ) and Operability ( $p = 0.016$ ). Post-hoc Wilcoxon signed-rank tests revealed significant differences in Enjoyment between con and fre conditions ( $p = 0.00035$ ) and between han and fre conditions ( $p = 2.4 \times 10^{-4}$ ). Additionally, a significant difference was detected in Operability between han and fre conditions ( $p = 0.0071$ ).

### 4 DISCUSSION

The results of this study provide evidence supporting both hypotheses. Regarding H1, the findings show that variations in contact plane tilt significantly affect visuo-haptic coherency. This was evident in the significant differences observed in subjective evaluations and statistical analysis of coherency across different methods.

The alignment of tactile and visual stimuli achieved through fingertip contact plane manipulation underscores the importance of surface inclination in shape perception, consistent with previous research [4].

For H2, manipulating fingertip contact planes significantly improved the quality of VR grasping experiences. Participants reported higher levels of enjoyment and operability when using the device compared to conventional methods. This suggests that the device not only enhances the plausibility of tactile feedback but also contributes to a more engaging and intuitive interaction in VR environments.

However, certain limitations must be addressed in future work. The current study focused on static tactile stimuli, which may not fully capture the dynamic nature of real-world interactions. Integrating real-time adjustments to contact plane angles based on user input could simulate properties such as softness or elasticity, broadening the scope of applications.

Moreover, while the compact design of the device is advantageous, long-term usability studies are needed to assess its ergonomic impact. Exploring alternative materials and configurations could further enhance user comfort and reduce device weight. Finally, expanding the range of shapes and including larger, more diverse participant samples will ensure broader applicability and validity of the findings.

### 5 CONCLUSION

This study demonstrates that fingertip contact plane manipulation is a promising approach to improving VR haptic interfaces. By enhancing visuo-haptic coherency and providing more plausible tactile feedback, this method contributes to more immersive and realistic VR experiences. The findings highlight the potential of this approach in applications such as gaming, training simulations, and virtual prototyping.

Future work should focus on refining the device's design to support dynamic stimuli, integrating visual retargeting techniques, and conducting longitudinal studies to assess its impact on user engagement and fatigue. We anticipate that continued advancements in this area will pave the way for more natural and intuitive haptic interactions in VR environments.

### ACKNOWLEDGMENTS

This work was supported in part by a grant from NAIST Granite Proram.

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